

James Ruse Agricultural High School 2010 Year 12 Mathematics Trial Exam

Question 1.	Marks
(a) Evaluate to 2 significant figures : $\frac{3.72 \times 1.96 + \sqrt{4.3 + 2.7^2}}{3.6 \times 1.8 + 3.1^3}$	1
(b) Rationalise the denominator and write in the form $a + b\sqrt{2}$: $\frac{3\sqrt{2}+4}{2\sqrt{2}-3}$ where a, b are real.	2
(c) Find the acute angle (to the nearest minute) that the line $4x - 11y + 9 = 0$ makes with the x axis.	2
(d) Graph $y = 2\sin 3x$ in the domain $-\pi \leq x \leq \pi$.	2
(e) Find $\lim_{h \rightarrow 0} \left(\frac{4^h - 1}{2^h - 1} \right)$	2
(f) Solve : $ x - 3 = 4x + 2$	3
Question 2.	
Three points A,B and C lie on the x-y plane. The lines l and k represent the lines AB and AC respectively. The equations of lines l and k are respectively: $3x - 4y - 100 = 0$ and $16x - 63y + 175 = 0$ respectively.	
(a) Show that $B(8, -19)$ lies on the line l .	1
(b) Find the co-ordinates A of the intersection of lines l and k .	3
(c) Find in general form the equation of the line m perpendicular to line l passing through B.	2
(d) Show that line m intersects line k at the point $C(-7, 1)$.	2
(e) Find the exact perpendicular distance of B from AC.	2
(f) Find the area of triangle ABC.	2

Question 3.	Marks
(a) Differentiate : (i) $\frac{3}{\sqrt{1-2x}}$	2
(ii) $\frac{\sin x}{x}$	2
(iii) $e^{\tan x}$	2
(b) Find (i) $\int \sqrt{e^{2x}} dx$	1
(ii) $\int (\cot x - \operatorname{cosec}^2 x) dx$	2
(c) Find in simplest terms : $\frac{d}{dx} \{x^2(2 \ln x - 1)\}$, hence evaluate $\int_1^e x \ln x dx$.	3

Question 4.

(a) Given the equation $x^2 = 16(y + 4)$	
(i) State the co-ordinates of the vertex.	1
(ii) Find the focal length	1
(iii) State the co-ordinates of the focus	1
(iv) Find in general form the equation of the tangent at $(-12, 5)$	2
(v) Find the co-ordinates of the point where the tangent meets the directrix.	1
(b) A jar has 15 red discs and 9 black discs, while another jar has 20 red discs, 15 black discs and 10 white discs. A disc is drawn from each jar. Find the probability of drawing discs of the same colour ?	2
(c) A car tyre of diameter 60cm is in contact with the road at the point P . After the car has travelled 1000km how high (to the nearest millimetre) is the point P from the ground.	4

Question 5.

(a) Given $N = x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \dots + xy^{n-2} + y^{n-1}$	
(i) Simplify N in terms of x and y .	2
(ii) Hence prove $11^{21} - 5^{21}$ is divisible by 3.	2
(b) Use Simpson's Rule with 3 function values to evaluate to 2 decimal places : $\int_0^2 \frac{4 dx}{2 \sin x + 1}$	3
(c) Solve to 2 decimal places : $3^{2x+1} - 3^x = 10$	3
(d) If the quadratic equation : $(k^2 + l^2)x^2 + 2l(k + m)x + l^2 + m^2 = 0$ has equal roots then show $l^2 = km$.	2

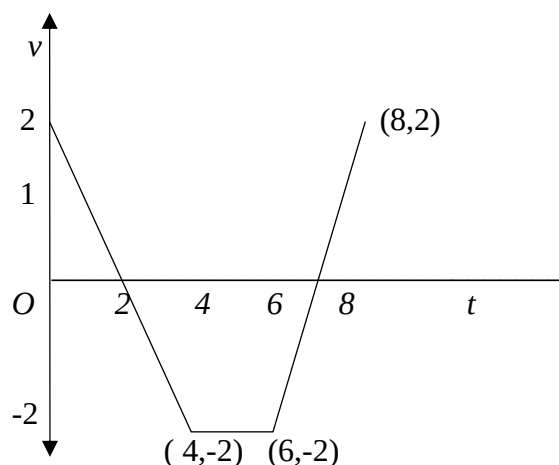
Question 6.**Marks**

- (a) The region bounded by the curve $y = x(6 - x)$ and $y = 8$ is rotated around the x axis.

4

Find the exact value of the Volume of revolution.

(b)



A particle of mass 2 kg moves in a straight line with velocity v m/s and displacement x m at time t seconds.

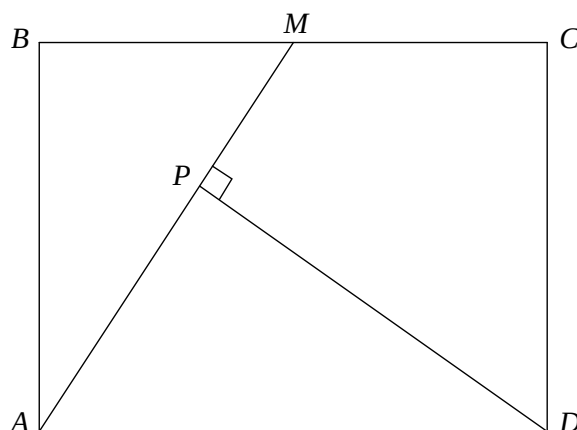
- (i) Graph acceleration $\ddot{x}$ m/s² versus time t seconds. 2
- (ii) Find the total distance travelled during the motion. 1
- (c) Find in general form the equation of the inflexional tangent on the curve : 5
- $$y = 15 + 12x + 6x^2 - 2x^3$$

Question 7.

- (a) (i) On the same axes graph : 4
- (α) the line $y = 1 - 2x$ showing x and y intercepts.
- (β) the curve $y = 5 - 2x - x^2$,
showing the co-ordinates of the vertex and y intercept only.
- (ii) Find the x values of the points A and B of the intersection of 2
- the line $y = 1 - 2x$ and the curve $y = 5 - 2x - x^2$.
- (iii) Evaluate the enclosed area between the line $y = 1 - 2x$ and 3
- the curve $y = 5 - 2x - x^2$.
- (b) The rate of decay $\frac{dM}{dt}$ of a radioactive substance is proportional to the mass M 3
- present.
- If it takes 51 minutes to decay to $\frac{1}{10}$ of its original mass find the half-life of the substance (nearest minute).

Question 8.**Marks**

(a)



$ABCD$ is a rectangle in which $AB=40\text{cm}$ and $AD=60\text{ cm}$.
 M is the midpoint of BC and DP is perpendicular to AM .

Draw a neat sketch of the above diagram.

- | | | |
|-------|--|---|
| (i) | Prove that triangles ABM and APD are similar. | 2 |
| (ii) | Calculate the length of PD . | 2 |
| (iii) | Show that the length of AP is 36 cm. Give reasons. | 2 |
| (iv) | Find the area of the quadrilateral $PMCD$. | 3 |
- (b) A plane flies from town O to town A , 275 km on a bearing of $032^{\circ}T$, then to town B 572 km on a bearing of $S\ 26^{\circ}E$. 3
- (i) Draw a diagram to show the above information.
- (ii) Find the final distance (nearest km), and bearing (nearest degree) from O .

Question 9.

- (a) A particle of mass m kg moves in a straight line with velocity v m/s and displacement x metres at time t seconds.

The velocity of the particle is given by : $v = 3\sqrt{1 + 9t}$.

- | | | |
|----------|---|---|
| Find (i) | the acceleration $\ddot{x}$ in terms of time t . | 1 |
| (ii) | the displacement of the particle as a function of time t if the particle is initially 1 metre to the left of the origin. | 2 |

		Marks
(b)	A man buys a house and land for \$500 000. He pays 20% deposit, and takes a loan for the remainder.	
(i)	Find the value of the deposit.	1
(ii)	If the loan is for 20 years, and the interest rate is 8% p.a. monthly reducible show that the amount owing after the first monthly repayment R is : $\$ (400\,000(\frac{151}{150}) - R)$	1
(iii)	Find the amount owing after n months.	2
(iv)	Find the monthly repayment.	2
(v)	Find the amount owing after the 144 th payment.	1
(vi)	The value of the land was originally valued at \$270 000 . If the value of the land was compounded yearly at 6% p.a. find the value of the land after the 144 th payment.	1
(vii)	After the 144 th payment an earthquake destroys the house. The insurance policy does not cover earthquakes. Could the man sell the land to pay the remainder of the loan? Give reasons.	1

Question 10.

A series S is given by :

$$S = x + \frac{2x^2}{x+1} + \frac{4x^3}{(x+1)^2} + \frac{8x^4}{(x+1)^3} + \dots$$

- | | | |
|-----|--|---|
| (a) | Sketch the curve $y = \frac{2x}{x+1}$, showing all asymptotes and intercepts with the axes. | 2 |
| (b) | Find the values of x for the sum to infinity to exist. | 2 |
| (c) | Show that the sum to infinity is given by :
$S_{\infty} = \frac{x^2+x}{1-x}$ | 2 |
| (d) | Show that $\frac{dS_{\infty}}{dx} = \frac{-x^2+2x+1}{(1-x)^2}$ | 2 |
| (e) | Find the minimum value of the sum to infinity.
Justify your answer. | 4 |

End of Exam

24 MATHEMATICS: Question 1...

Suggested Solutions

Marks

Marker's Comments

(a)
$$\frac{3.72 \times 1.96 + \sqrt{4.3 + 2.7^2}}{3.6 \times 1.8 + 3.13}$$

$$= 0.29488$$

$$= 0.29 \quad (2 \text{ sig figures})$$

$\frac{1}{2}$

$\frac{1}{2}$

If they wrote out the calculator display and it was wrong but they rounded up correctly they get $\frac{1}{2}$ mk.

(b)
$$\frac{(3\sqrt{2}+4)(2\sqrt{2}+3)}{(2\sqrt{2}-3)(2\sqrt{2}+3)} = \frac{12+9\sqrt{2}+8\sqrt{2}+12}{8-9}$$

$$= -24 - 17\sqrt{2}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

(c)
$$m = \frac{4}{11}$$

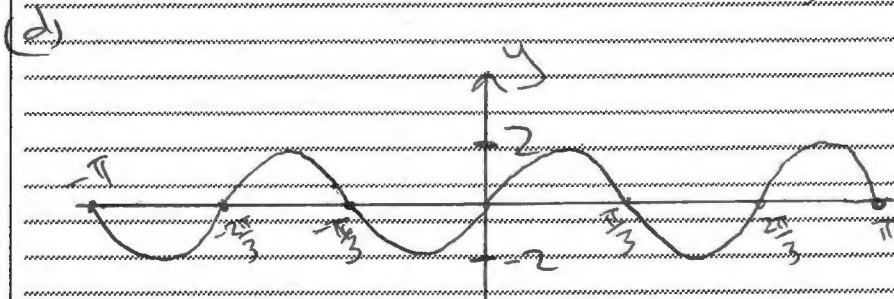
$$\tan \theta = \frac{4}{11}$$

$$\theta = 19^\circ 51' \quad (\text{nearest minute})$$

1

$\frac{1}{2}$

$\frac{1}{2}$



$\frac{1}{2}$ - intercepts

$\frac{1}{2}$ - shape

$\frac{1}{2}$ - amplitude

$\frac{1}{2}$ - period

(e)
$$\lim_{h \rightarrow 0} \frac{(4^h - 1)}{2^h - 1} = \lim_{h \rightarrow 0} \frac{(2^h - 1)(2^h + 1)}{2^h - 1}$$

$$= \lim_{h \rightarrow 0} (2^h + 1)$$

$$= 1 + 1$$

$$= 2$$

1

$\frac{1}{2}$

$\frac{1}{2}$

an answer of "2" with no working = $\frac{1}{2}$ mk.

(f)

$$4x + 2 = 3 - x$$

$$5x = 1$$

$$x = \frac{1}{5}$$

2 mks

1 mk

* scored 2 mks if you solved algebraically but forgot to check the answers.

2010 TRIAL 2U MATHEMATICS: Question...2...

Suggested Solutions

Marks

Marker's Comments

2(a) Show B(8, -19) lies on $3x - 4y - 100 = 0$
 $LHS = 3 \times 8 - 4 \times -19 - 100$
 $= 24 + 76 - 100$
 $= 0$
 $\therefore LHS = RHS$, hence B(8, -19) lies on ℓ .

1

$0 = 0$ without saying $LHS = RHS$ lost $\frac{1}{2}$ mark

(b) $\ell: 3x - 4y - 100 = 0 \dots (1)$
 $k: 16x - 63y + 175 = 0 \dots (2)$
 $(1) \times 16$ and $(2) \times 3$
 $48x - 64y = 1600 \dots (3)$
 $48x - 189y = -525 \dots (4)$
 $(3) - (4)$
 $125y = 2125$
 $y = 17$
 Subst. into (1)
 $3x - 68 = 100$
 $3x = 168$
 $x = 56$
 $\therefore A(56, 17)$

1

1

1 mark deducted for each error

(c) $m_\ell = \frac{3}{4}$
 Gradient of a line perpendicular to ℓ is $-\frac{4}{3}$ as $m_1 \times m_2 = -1$ for perp. lines
 Eqn is: $y + 19 = -\frac{4}{3}(x - 8)$
 $3y + 57 = -4x + 32$
 $\therefore$ Eqn. of the line in general form is:
 $4x + 3y + 25 = 0$

1

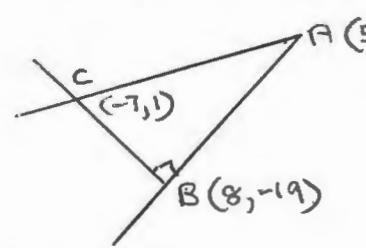
1

Not in general form, max of 1 mark only

(d) For C
 $4x + 3y = -25 \dots (1)$
 $16x - 63y = -175 \dots (2)$
 $(1) \times 4$
 $16x + 12y = -100 \dots (3)$
 $(3) - (2)$
 $-75y = 75$
 $y = -1 \dots (4)$
 Sub (4) in (1)
 $4x + 3(-1) = -25$
 $4x - 3 = -25$
 $4x = -22$
 $x = -\frac{11}{2}$
 $\therefore C(-\frac{11}{2}, -1)$

1

1

Suggested Solutions	Marks	Marker's Comments
2(e) $D = \frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}}$ $= \frac{16 \times 8 - 63 \times -19 + 175}{\sqrt{16^2 + 63^2}}$ $= \frac{1500}{65} = \frac{300}{13}$ $\therefore D = 23\frac{1}{13}$ units	1 1	No penalty for units missing
(f)  $A(56, 17)$ $B(8, -19)$ $C(-7, 1)$ $AC = \sqrt{63^2 + 16^2}$ $= 65$ units $\therefore \text{Area } \triangle ABC = \frac{1}{2} \times 65 \times 23\frac{1}{13}$ $A = 750$ units²	1 1	

Question 3

a) Differentiate

$$i) \frac{3}{\sqrt{1-2x}} = \frac{d}{dx} 3(1-2x)^{-\frac{1}{2}} = 3 \times -\frac{1}{2} \cdot -2(1-2x)^{-\frac{3}{2}} \checkmark$$

$$= 3(1-2x)^{-\frac{3}{2}} \checkmark$$

OR $\frac{3}{(1-2x)^{\frac{3}{2}}}$

or $\frac{3}{(1-2x)(\sqrt{1-2x})}$

$\frac{-3}{2\sqrt{(1-2x)^3}}$ earned 1 mark

$\frac{-3}{\sqrt{1-2x}}$ earned 1 mark

2 marks

$$ii) \frac{d}{dx} \frac{\sin x}{x} = \frac{x \cos x - \sin x}{x^2} \checkmark$$

1 1/2 mark if $\frac{\sin x - x \cos x}{x^2}$ 2 marks

$$iii) \frac{d}{dx} e^{\tan x} = \sec^2 x \cdot e^{\tan x} \checkmark$$

2 marks

$$b) i) \int \sqrt{e^{2x}} dx = \int e^x dx$$

$$= e^x + C \checkmark$$

1 mark

-1/2 if no C
Note cannot apply chain rule for $\int$.

$$ii) \int (\cot x - \operatorname{cosec}^2 x) dx = \int \left(\frac{\cos x}{\sin x} - \operatorname{cosec}^2 x \right) dx$$

-1/2 wrong sign

$$\int \frac{\cos x}{\sin x} dx = \ln \sin x + C \checkmark$$

We know the derivative of $\cot x = -\operatorname{cosec}^2 x$
proof $\cot x = \frac{\cos x}{\sin x}$ let $u = \cos x \rightarrow u' = -\sin x$
 $v = \sin x \rightarrow v' = \cos x$

Using Quotient Rule $\frac{\sin x(-\sin x) - \cos^2 x}{\sin^2 x}$

$$= \frac{-1(\sin^2 x + \cos^2 x)}{\sin^2 x}$$

$$= \frac{-1}{\sin^2 x}$$

$$= -\operatorname{cosec}^2 x$$

$$\therefore \int -\operatorname{cosec}^2 x \cdot dx = \cot x + C \checkmark$$

$$\therefore \int \cot x - \operatorname{cosec}^2 x = \ln \sin x + \cot x + C$$

2 marks

$$c) \frac{d}{dx} x^2 (2 \ln x - 1) = x^2 \cdot \frac{2}{x} + (2 \ln x - 1) 2x$$

Use product rule

$$= 4x \ln x$$

$$\int_1^e x \ln x dx = \left\{ \frac{1}{4} x^2 [2 \ln x - 1] \right\}_1^e \checkmark$$

$$= \frac{1}{4} [e^2 (2-1) - 1 \cdot (-1)]$$

$$= \frac{e^2 + 1}{4} \checkmark$$

3 marks

Question 4

a) (i) Vertex $(0, -4)$ ✓

1 mark

ii) $4a = 16$
 $a = 4$ ✓

1 mark

iii) focus $(0, 0)$ ✓

1 mark

iv) $y = \frac{x^2}{16} - 4$

$\frac{dy}{dx} = \frac{x}{8}$

at $x = -12$

$m = \frac{-12}{8}$

$m = -\frac{3}{2}$ ✓

Equation of tangent

$y - y_1 = m(x - x_1)$

$y - 5 = -\frac{3}{2}[x + 12]$

$2y - 10 = 3x - 36$

$3x + 2y + 26 = 0$ ✓

few did not

simplify

$\frac{-12}{8} = -\frac{3}{2}$

2 marks

v) Directrix $y = -8$

$3x + 2(-8) + 26 = 0$

$3x = -10$

$x = -3\frac{1}{3}$

∴ point is $(-3\frac{1}{3}, -8)$ ✓

1 mark

b) $P(R, R) + P(B, B) = \frac{15}{24} \times \frac{20}{45} + \frac{9}{24} \times \frac{15}{45}$

$= \frac{5}{8} \times \frac{4}{9} + \frac{3}{8} \times \frac{3}{9}$ ✓

$= \frac{29}{72}$ ✓

2 marks

c) Circumference of Tyre $= 60\pi$

Distance travelled $= 1000 \text{ km} = 1000000 \text{ m}$
 $= 100000000 \text{ cm}$
 $= 1000000000 \text{ mm}$

Number of revolutions $= \frac{\text{Distance travelled}}{\text{circumference tyre}}$
 $= \frac{1000000000 \text{ cm}}{60\pi \text{ cm}}$
 $= 530516.477 \text{ rev.}$

There is 530,516 complete revolutions with 0.477 of a revolution left over

Hence 0.477 of circumference ($60\pi \text{ cm}$) is left over

i.e. $0.477 \times 60\pi$

$= 89.91238175 \text{ cm.}$

calculate θ

$l = r\theta$

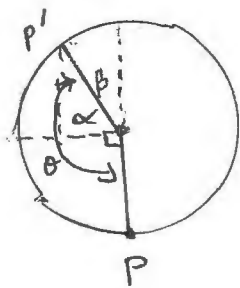
$89.91238175 = 30\theta$

$\theta = 2.997079592^\circ$ or 0.954π
or 171.72°

* few $r = 60 \text{ cm}$ instead of 30 cm .

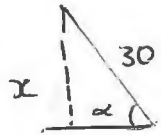
* Did not change both distance and radius into same unit

4 marks



$$\alpha = \theta - \frac{\pi}{2}$$

$$= 1.426283065$$



$$\sin \alpha = \frac{x}{30}$$

$$x = 30 \sin \alpha$$

$$\approx 29.8$$

$$\therefore \text{Total height} = 30 + x$$

$$= 30 + 29.6865804$$

$$= 59.6865804$$

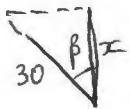
$$= 59.7 \text{ mm}$$

OR

$$\beta = \pi - \theta$$

$$= \pi - 1.7172$$

$$= 4.28$$



$$\cos 8.28 = \frac{x}{30}$$

$$x = 30 \cos 8.28$$

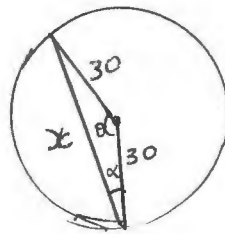
$$= 29.68728356$$

$$\therefore \text{height } 29.68728356$$

$$+ 30$$

$$= 59.68728356$$

$$\text{height} = 59.7 \text{ mm}$$



$$\alpha = \frac{180 - 171.72}{2}$$

$$= 4.14^\circ$$

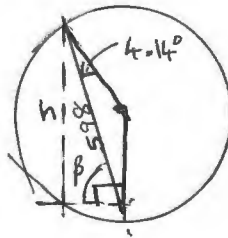
Sine Rule

$$\frac{x}{\sin 171.72^\circ} = \frac{30}{\sin 4.14^\circ}$$

$$x = \frac{30 \sin 171.72^\circ}{\sin 4.14^\circ}$$

$$= 59.843 \text{ cm}$$

$$= 59.8 \text{ mm}$$



$$\beta = 90 - 4.14^\circ$$

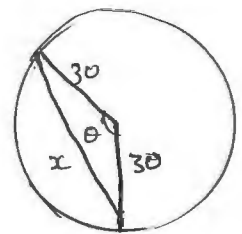
$$= 85.86$$

$$\sin \beta = \frac{h}{59.8}$$

$$h = 59.8 \sin 85.86$$

$$h = 59.64395939$$

$$h = 59.6 \text{ mm}$$



cosine rule

$$x^2 = 30^2 + 30^2 - 2(30)(30) \cos 0.984 \pi$$

$$= 3581.237013$$

$$x = 59.843$$

$$= 59.8 \text{ mm}$$

similarity

as in

previous
example

Q 5

a)
$$N = \frac{x^{n-1} \left[\left(\frac{y}{x} \right)^n - 1 \right]}{\frac{y}{x} - 1} \quad 1m$$

$$N = \frac{x^n - y^n}{x - y} \quad 1m$$

$$\begin{aligned} x^n - y^n &= (x - y)N \\ 11^{21} - 5^{21} &= (11 - 5) \left[11^{20} + 11^{19} \cdot 5 + \dots + 5^{20} \right] \quad 1m \\ &= 6 \times (11^{20} + 11^{19} \cdot 5 + \dots + 5^{20}) \\ &= 3 \times 2 \times (11^{20} + 11^{19} \cdot 5 + \dots + 5^{20}) \end{aligned}$$

$\therefore 11^{21} - 5^{21}$ is divisible by 3 ^{integer} $\neq$ $1m$

b)

x	0	1	2
$\frac{4}{2\sin x + 1}$	4	1.4909	1.4191

1m

$$\begin{aligned} \int_0^2 \frac{4dx}{2\sin x + 1} &= \frac{2-0}{6} \left[4 + 4 \left[\frac{4}{2\sin 1 + 1} \right] + \frac{4}{2\sin 2 + 1} \right] \quad 1m \\ &= 3.79425 \\ &= \underline{\underline{3.79 \text{ (2dp)}}} \quad 1m \end{aligned}$$

c)
$$3 \cdot 3^{2x} - 3^x - 10 = 0$$

Put $u = 3^x$

$$3u^2 - u - 10 = 0 \quad 1m$$

$$(3u + 5)(u - 2) = 0$$

$$u = 2 \quad \text{or} \quad u = -5/3$$

$$\therefore 3^x = 2 \quad \text{or} \quad 3^x = -5/3 \quad 1m$$

but $3^x > 0 \therefore 3^x = 2$ only

$$x = \frac{\ln 2}{\ln 3} = 0.6309 \dots$$

$$x = \underline{\underline{0.63 \text{ (2dp)}}} \quad 1m$$

many students did not simplify completely to lowest term.

many students did not justify $11^{20} + 11^{19} \cdot 5 + \dots + 5^{20}$ is an integer and simply say N is an integer for 1m only.

2h Math Trial 2010

5 d)

$\Delta = 0$ for equal roots

$$[2l(k+m)]^2 - 4[k^2 + l^2][l^2 + m^2] = 0 \quad | : m$$

$$4l^2[k^2 + 2km + m^2] - 4[\cancel{k^2}l^2 + k^2m^2 + l^4 + \cancel{l^2m^2}] = 0$$

$$\cancel{4l^2} \cdot 2km - \cancel{4k^2}m^2 - \cancel{4l^4} = 0$$

$$l^4 + k^2m^2 - 2kml^2 = 0$$

$$(l^2 - km)^2 = 0 \quad | : m$$

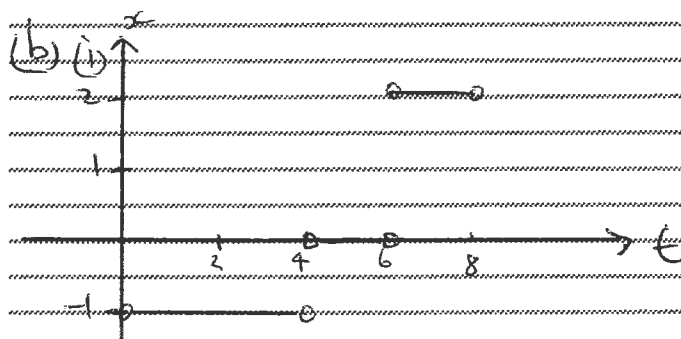
$$l^2 = km \quad \#$$

many students
made mistakes
& cannot
complete square
next 1m

24 **MATHEMATICS: Question 6** ...
Suggested Solutions

(a) Intersection points: $x(6-x)=8$
 $x^2-6x+8=0$
 $(x-4)(x-2)=0$
 $x=4$ or $x=2$

$$\begin{aligned} \text{Volume} &= \pi \int_2^4 x^2(6-x) dx - \pi \int_2^4 8^2 dx \\ &= \pi \int_2^4 (36x^2 - 12x^3 + x^4) dx - \pi \int_2^4 64 dx \\ &= \pi \left[36x^3 - 12x^4 + \frac{x^5}{5} - 64x \right]_2^4 \\ &= \pi \left[\frac{4^5}{5} + 12(4)^3 - 3(4)^4 - (4(4)) \right] - \pi \left[12 \times 8 - 3(16) + \frac{32}{5} - 128 \right] \\ &= \pi (204\frac{4}{5} - 768 + 768 - 256) - \pi (96 - 48 + 6\frac{4}{5} - 128) \\ &= \pi (198\frac{4}{5} - 176) \\ &= 22\frac{2}{5}\pi \text{ or } \frac{112\pi}{5} \text{ units}^3 \text{ or } 22.4\pi \text{ units}^3 \end{aligned}$$



(ii) Total distance = $\frac{2 \times 2}{2} + \frac{2 \times 2}{2} + 2 \times 2 + \frac{1 \times 2}{2} + \frac{1 \times 2}{2}$
 $= 2 + 2 + 4 + 1 + 1$
 $= 10 \text{ metres}$

(c) $y = 15 + 12x + 6x^2 - 2x^3$
 $\frac{dy}{dx} = 12 + 12x - 6x^2$

Marks

Marker's Comments

1

If they had the wrong limits, 1 mark off

1/2

* 1/2 mk off if they forgot to square the fns.

* 1/2 mk off if the fns are around the wrong way

* 1/2 mk off if squared as one big function

* 1/2 mk off for every calculator error

* 1 mk off if they forgot $\pi \int_2^4 8^2 dx$.

1

1/2

2 mks

* 1/2 mk off for no open circles

* 1/2 mk off if lines joined up

* 0 for curves!

1

* 1/2 mk if they got 8 metres.

1/2

MATHEMATICS: Question...6... continued

Suggested Solutions

Marks

Marker's Comments

$$\frac{d^2y}{dx^2} = 12 - 12x$$

possible points of inflexion when $\frac{d^2y}{dx^2} = 0$

$$12 - 12x = 0$$

$$x = 1$$

when $x = 1$, $y = 15 + 12 + 6 - 2$

$$y = 31 \quad (1, 31)$$

when $x = 1$, $y' = 12 + 12 - 6$
 $= 18$
 $\therefore m = 18$

test for change in concavity

x	0.9	1	1.1
$\frac{d^2y}{dx^2}$	1.2	0	-1.2
concavity	up	-	down

change in concavity
at $x = 1$

$(1, 31)$ is a
point of inflexion

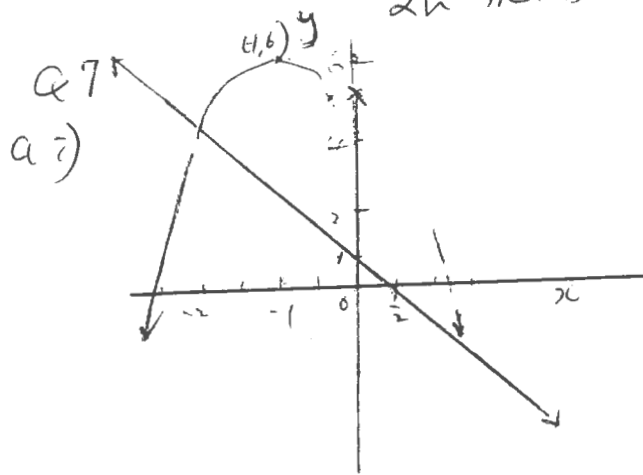
eqn. of inflexional tangent is

$$y - y_1 = m(x - x_1)$$

$$y - 31 = 18(x - 1)$$

$$18x - y + 13 = 0$$

* 1/2 mk if they
didn't use numbers
or if they didn't
state there's a
change in concavity



intercepts $\frac{1}{2}$ m each
 $(0, 1), (\frac{1}{2}, 0), (0, 1)$
 vertex $(-1, 6)$ 1 m
 parabola 1 m
 straight line $\frac{1}{2}$ m

Comments

generally well done.

ii) intersect at $x = \pm 2$ 2 m

iii) Area = $\int_{-2}^1 (5 - 2x - x^2) - (1 - 2x) dx$ 1 m

$$= \left[4x - \frac{x^3}{3} \right]_{-2}^1$$

1 m

$$= 10 \frac{2}{3} \text{ unit}^2$$

1 m

generally well done

b)

$$M = Ae^{-kt}$$

1 m

$$k = \frac{\ln 10}{51}$$

1 m

$$\frac{A}{2} = A e^{-t \cdot \frac{\ln 10}{51}}$$

1 m

$$t = 15.35$$

$\frac{1}{2}$ m

$$= 15 \text{ minutes (nearest min)}$$

$\frac{1}{2}$ m

A few students wrote

$$k = -\frac{\ln \frac{9}{10}}{51}$$

and

$$t = 336 \text{ min}$$

get 2 m

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(i) In Δ s ABM, APD

$$\angle B = \angle P \quad (\text{both } 90^\circ)$$

$$\angle BMA = \angle PAD \quad (\text{alternate angles are equal as } BM \parallel AD)$$

$\therefore \Delta ABM \parallel \Delta APD$ (equiangular)

(ii) $\frac{PD}{AB} = \frac{AD}{AM}$ (corresponding sides in similar triangles are in the same ratio)

$$\frac{PD}{40} = \frac{60}{50}$$

$$PD = \frac{60}{50} \times 40$$

$$PD = 48 \text{ cm}$$

$$\begin{aligned} AB^2 + BM^2 &= AM^2 \\ 30^2 + 40^2 &= AM^2 \\ AM &= 50 \\ (\text{by Pythagoras}) \end{aligned}$$

(iii) $AP^2 + PD^2 = AD^2$ (by Pythagoras)

$$AP^2 + 48^2 = 60^2$$

$$AP^2 = 1296$$

$$AP = 36 \text{ cm}$$

(iv) $\text{Area}_{PMCD} = A_{ACD} - A_{ABM} - A_{APD}$

$$= (60 \times 40) - \left(\frac{40 \times 30}{2} \right) - \left(\frac{48 \times 36}{2} \right)$$

$$= 936 \text{ units}^2$$

$$\begin{aligned} \text{or } PM &= AM - PA \\ &= 50 - 36 \\ &= 14 \end{aligned}$$

$$\text{Area}_{DCM} = \frac{1}{2} \times 30 \times 40 = 600$$

$$\text{Area}_{PMD} = \frac{1}{2} \times 14 \times 48 = 336$$

$$\therefore \text{Area}_{PMCD} = \text{Area}_{DCM} + \text{Area}_{PMD}$$

$$= 600 + 336$$

$$= 936 \text{ units}^2$$

-1/2 if ratio is written incorrectly.

(or could have done it using ratio of corresponding sides in similar triangles)

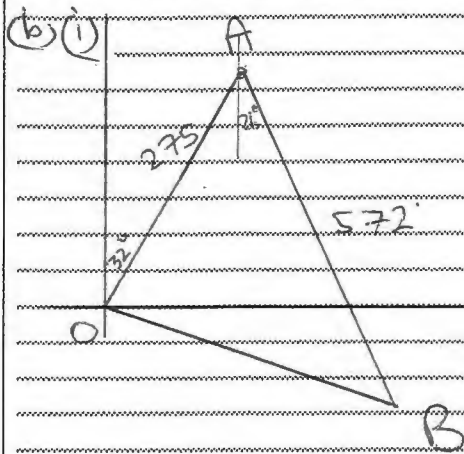
(1/2 off for each error).

24 MATHEMATICS: Question 8 ..

Suggested Solutions

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Marker's Comments



$\frac{1}{2}$

if all of the information is correct!

(ii) $OB^2 = 275^2 + 572^2 - 2(275)(572)\cos 58^\circ$
 $= 402809 - 314600\cos 58^\circ$
 $= 402809 - 166712.6005$
 $= 236096.3995$
 $\therefore OB = 485.8975193$
 $OB = 486 \text{ km (nearest km)}$

$\frac{1}{2}$

$\frac{1}{2}$

$572^2 = 275^2 + 485.89^2 - 2(275)(485.89)\cos AOB$
 $\therefore \cos AOB = \frac{275^2 + 485.89^2 - 572^2}{2(275)(485.89)}$
 $AOB = 93.19^\circ$

$\left. \begin{array}{l} \frac{1}{2} \\ \frac{1}{2} \end{array} \right\} 1$

* If they used the sine rule and didn't find the obtuse angle, they lose $\frac{1}{2}$ mk + lose $\frac{1}{2}$ mk from bearing ie +1 off the total.

$\therefore \text{Bearing is } (93^\circ + 32^\circ)^\circ$
 $= 125^\circ \text{ (nearest degree)}$

$\frac{1}{2}$

Question 9.

a(i) $\ddot{x} = \frac{dv}{dt}$
 $= \frac{d}{dt} 3[1+9t]^{\frac{1}{2}}$
 $= \frac{3 \cdot 9}{2} (1+9t)^{-\frac{1}{2}}$
 $\ddot{x} = \frac{27}{2\sqrt{1+9t}}$ ✓

[1 mark]

Do not use
decimals

$$\frac{13.5}{\sqrt{1+9t}}$$

ii) $x = \int 3(1+9t)^{\frac{1}{2}} dt$
 $x = \frac{2}{9} (1+9t)^{\frac{3}{2}} + C$ ✓

when $t=0, x=-1$

$$-1 = \frac{2}{9} + C$$

$$C = -\frac{11}{9}$$

$\therefore$ Displacement $x = \frac{2}{9} (1+9t)^{\frac{3}{2}} - \frac{11}{9}$ ✓

[2 marks]

b(i) Deposit = 20% of \$500000
 $= \$100000$ ✓

[1 mark]

ii) loan for \$400000
 Monthly interest = $\frac{8}{12}\%$ = $\frac{8}{1200} = \frac{1}{150}$

Amount owing after 1st month.

$$\$400000 \times \left(1 + \frac{8}{1200}\right) - R$$

$$= 400000 \left(\frac{1+150}{150}\right) - R$$

$$= 400000 \left(\frac{151}{150}\right) - R$$

[1 mark] ✓

Had to show
 $\frac{8}{12}\%$ to gain
a mark.

iii) $A_2 = 400000 \left(\frac{151}{150} - R\right) \frac{151}{150} - R$
 $= 400000 \left(\frac{151}{150}\right)^2 - R \left(1 + \frac{151}{150}\right)$

$$A_3 = \left[400000 \left(\frac{151}{150}\right)^2 - R \left(1 + \frac{151}{150}\right) \right] \frac{151}{150} - R$$

$$= 400000 \left(\frac{151}{150}\right)^3 - R \left[1 + \frac{151}{150} + \left(\frac{151}{150}\right)^2 \right] - R$$

$\therefore$ Amount after n^{th} month
 $A_n = 400000 \left(\frac{151}{150}\right)^n - R \left[1 + \frac{151}{150} + \left(\frac{151}{150}\right)^2 + \dots + \left(\frac{151}{150}\right)^{n-1} \right]$ ✓ ✓

To show the answer
you need to
get a pattern
for at least
3 months.

[2 marks]

iv Find monthly repayment

$$A = 0 \quad \text{and} \quad n = 20 \times 12 = 240$$

$$0 = 400000 \left(\frac{151}{150} \right)^{240} - R \left[1 + \frac{151}{150} + \left(\frac{151}{150} \right)^2 + \dots + \left(\frac{151}{150} \right)^{239} \right]$$

$$R = \frac{400000 \left(\frac{151}{150} \right)^{240}}{\frac{\left(\frac{151}{150} \right)^{240} - 1}{\frac{151}{150} - 1}} \quad \checkmark$$

$$= \frac{400000 \left(\frac{151}{150} \right)^{240} \cdot \frac{1}{150}}{\left(\frac{151}{150} \right)^{240} - 1}$$

$$= \$3345.76 \quad \checkmark$$

2 marks

Round off to
2 decimal place

$$v) A_{144} = 400000 \left(\frac{151}{150} \right)^{144} - 3345.76 \left[\frac{\left(\frac{151}{150} \right)^{144} - 1}{\left(\frac{151}{150} - 1 \right)} \right]$$

$$= \$236672.36 \quad \checkmark$$

1 mark

vi) Value of land

$$\text{value} = 270000 (1.06)^{12}$$

$$= \$543293.05 \quad \checkmark$$

1 mark

vii) Man can sell land as

$$\$543293.05 > \$236672.36 \quad \checkmark$$

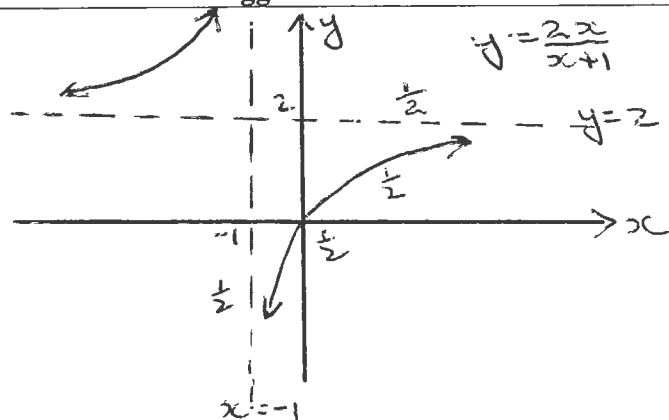
1 mark

Suggested Solutions

Marks

Marker's Comments

10 (a)



2

Shape $\frac{1}{2}$ mark
y intercept labelled
gained $\frac{1}{2}$ mark
except when there
were more than
2 mistakes on
the graph.

$$(b) \quad \left| \frac{2x}{1+x} \right| < 1$$

$$\frac{2x}{1+x} = 1$$

$$x = 1$$

$$\left(\frac{1}{2} \right)$$

or

$$\frac{2x}{1+x} = -1$$

$$3x = -1$$

$$x = -\frac{1}{3} \quad \left(\frac{1}{2} \right)$$

 $\therefore$ For sum to infinity

$$\left(\frac{1}{2} \right) \quad -\frac{1}{3} < x < 1$$

$$[n.b. x=0, S=0]$$

2

$$(c) \quad S_{\infty} = \frac{a}{1-r}$$

$$= \frac{2x}{1-\frac{2x}{1+x}}$$

$$= \frac{x(1+x)}{1+x-2x}$$

$$\therefore S_{\infty} = \frac{x^2+x}{1-x}$$

where $a=x$

$$r = \frac{2x}{1+x}$$

1

1

$$(d) \quad S = \frac{x^2+x}{1-x}$$

$$\frac{dS}{dx} = \frac{(1-x)(2x+1) - (x^2+x)(-1)}{(1-x)^2}$$

$$= \frac{2x+1-2x^2-x+x^2+x}{(1-x)^2}$$

$$\therefore \frac{dS}{dx} = \frac{-x^2+2x+1}{(1-x)^2}$$

1

1

Suggested Solutions

Marks

Marker's Comments

$$(e) \quad \frac{ds}{dx} = \frac{-x^2 + 2x + 1}{(1-x)^2}$$

For minimum $\frac{ds}{dx} = 0$

$$\therefore x^2 - 2x - 1 = 0$$

$$x = \frac{2 \pm \sqrt{4+4}}{2}$$

$$= 1 \pm \sqrt{2}$$

But $-\frac{1}{3} < x < 1$ from (b)

But $1 - \sqrt{2} < -\frac{1}{3}$ and $1 + \sqrt{2} > 1$ then

Since S_{∞} is continuous for $-\frac{1}{3} < x < 1$ then the minimum occurs at the endpoints.

Test end pts.

$$\text{Minimum} = \lim_{x \rightarrow -\frac{1}{3}} S = \frac{\left(-\frac{1}{3}\right)^2 - \frac{1}{3}}{1 + \frac{1}{3}} \quad \text{and} \quad \lim_{x \rightarrow 1} S \rightarrow \infty$$

$$= \frac{1-3}{4+3}$$

$$= -\frac{1}{4}$$

If the question was made easier by using any other domain except $-\frac{1}{3} < x < 1$, max of 2 marks, BUT needed to test endpoints of their domain to gain the 2 marks.

When $x = 1 - \sqrt{2}$ an answer of $2\sqrt{2} - 3$ ($\approx 0.1715 \dots$) with working received $1\frac{1}{2}$ as question had been greatly simplified

$$\text{i.e. when } x = 1 - \sqrt{2}$$

$$S_{\infty} = \frac{(1 - \sqrt{2})^2 + 1 - \sqrt{2}}{1 - (1 - \sqrt{2})}$$

$$= \frac{1 - 2\sqrt{2} + 2 + 1 - \sqrt{2}}{\sqrt{2}}$$

$$= \frac{4 - 3\sqrt{2}}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{4\sqrt{2} - 6}{2}$$

$$= 2\sqrt{2} - 3$$